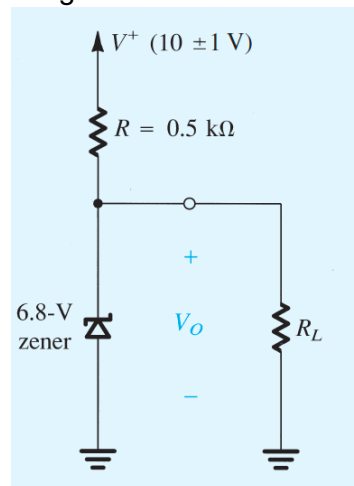
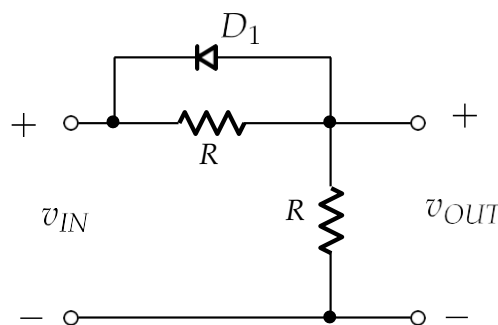


Homework 8 Topic: Diodes II + BJTs

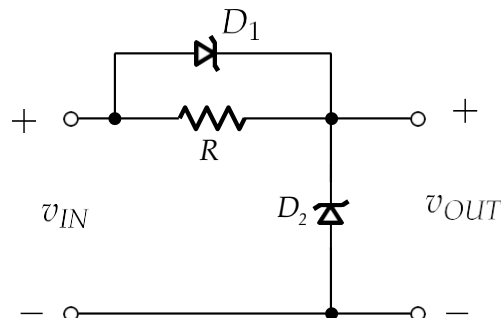
- Consider a half-wave peak rectifier fed with a voltage v_s having a triangular waveform with 24-V peak-to-peak amplitude, zero average, and 1-kHz frequency. Assume that the diode has a 0.7-V drop when conducting. Let the load resistance $R = 100\Omega$ and the filter capacitor $C = 100\mu\text{F}$. Find the average dc output voltage, the time interval during which the diode conducts, the average diode current during conduction, and the maximum diode current.
- Zener Diode Regulator: For this problem you will need the data sheet for a Zener Diode, which you can find your own or use the one Fairchild Semiconductor that is provided on Canvas. Note that the data sheet provides values for V_Z , which is measured at the current I_Z , as well as for the series resistance. We want to design a 12 V regulator using a reverse-bias Zener diode similar to the circuit below the load of $2.2\text{ k}\Omega$. For this design problem, the supply voltage V^+ is $25 \pm 2.5\text{ V}$ (not $10 \pm 1\text{ V}$ as shown in the figure below).
 - Choose one Zener diode to regulate the load voltage to approximately 12 V. Based on the information in the data sheet, find V_{Z0} for this diode.
 - If the Zener current is 25 mA, find V_Z . Then find the current through R , and from that determine a value for R . Use the nominal voltage for the power supply (i.e., 25 V).
 - Find the line regulation.



- For the circuit below using the constant voltage drop model with $V_{D0} = 0.7\text{ V}$ and $r_D = 0$:
 - Write an expression for v_{OUT} in terms of v_{IN} .
 - Sketch a graph of v_{OUT} vs. v_{IN} .



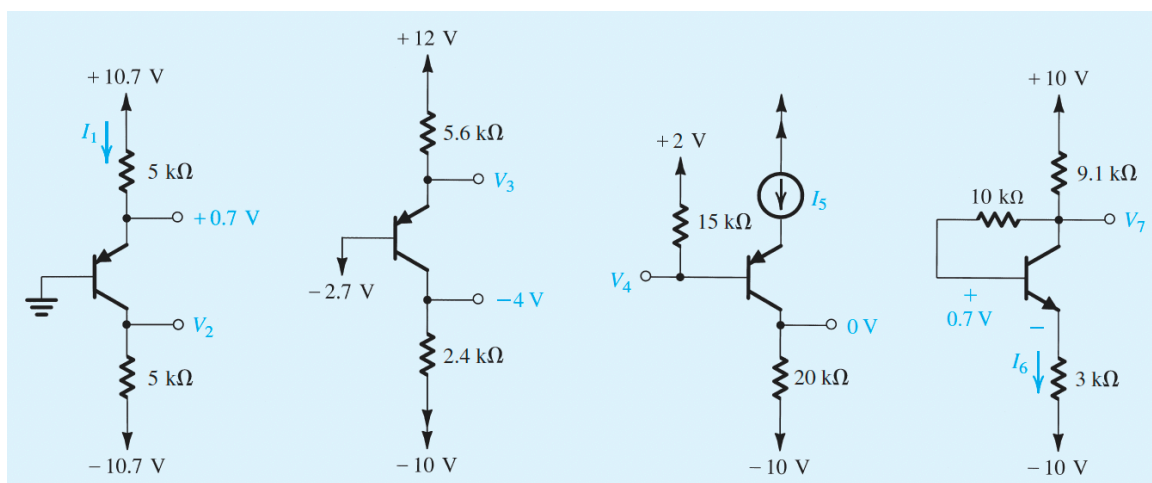
4. Repeat problem 3 with $r_D = R$.
5. For the circuit below using the constant voltage drop model with $V_{D0} = 0.7\text{ V}$, $V_{Z0} = 1.4\text{ V}$, $r_D = r_z = 0$:
 - a. Write an expression for v_{OUT} in terms of v_{IN} .
 - b. Sketch a graph of v_{OUT} vs. v_{IN} .



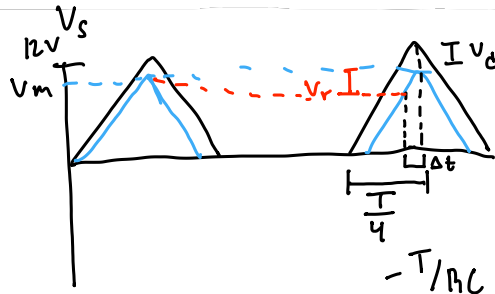
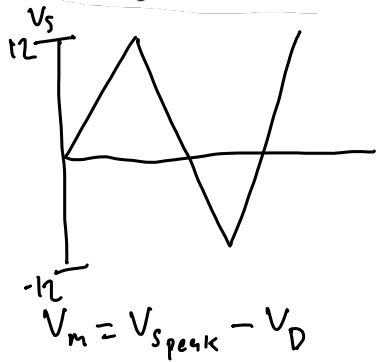
6. Measurements of V_{BE} and two terminal currents taken on a number of npn transistors operating in the active mode are tabulated below. For each, calculate the missing current value as well as α , β , and I_S as indicated by the table.

Transistor	a	b	c	d	e
V_{BE} (mV)	700	690	580	780	820
I_C (mA)	1.000	1.000		10.10	
I_B (μA)	10		5	120	1050
I_E (mA)		1.020	0.235		75.00
α					
β					
I_S					

7. For the circuits below, assume that the transistors have very large β . Some measurements have been made on these circuits, with the results indicated in the figure. Find the values of the other labeled voltages and currents.



1. Consider a half-wave peak rectifier fed with a voltage v_s having a triangular waveform with 24-V peak-to-peak amplitude, zero average, and 1-kHz frequency. Assume that the diode has a 0.7-V drop when conducting. Let the load resistance $R = 100\Omega$ and the filter capacitor $C = 100\mu\text{F}$. Find the average dc output voltage, the time interval during which the diode conducts, the average diode current during conduction, and the maximum diode current.



$$V_m = 12 - (0.7)$$

$$V_m = 11.3 \text{ [V]}$$

$$V_{dc,avg} \approx V_m - \frac{V_m}{2fRC}$$

$$V_{dc,avg} \approx 11.3 - \frac{11.3}{2(1k)(100)(100\mu)}$$

$$V_{dc,avg} = 10.735 \text{ [V]}$$

$$V_m - V_m e^{-T/RC} = V_r$$

$$V_m (1 - (1 - \frac{T}{RC})) \approx V_r$$

$$V_m \frac{T}{RC} \approx V_r$$

$$\frac{V_m}{fRC} \approx V_r$$

$$\frac{11.3}{1k(100)(100\mu)} \approx V_r$$

$$V_r = 1.13$$

$$RC = .01 > T = .001$$

$$\therefore e^{-T/RC} \approx 1 - \frac{T}{RC}$$

$$\frac{V_{speak}}{\frac{T}{4}} = \frac{V_r}{\Delta t}$$

$$4V_{sp} = \frac{V_r}{\Delta t}$$

$$\Delta t = \frac{1.13}{4(1k)(12)}$$

$$\Delta t = 23.5 \mu\text{s}$$

interval diode conducts

$$i_L = \frac{V_{dc}}{R}$$

$$i_L = C \frac{dV_c}{dt} \approx i_L \Delta t = C V_r$$

$$i_{Davg} = i_{Lavg} + i_L$$

$$i_{Davg} = C \frac{dV_c}{dt} + \frac{V_{dc}}{R}$$

$$i_{Davg} = \frac{C V_r}{\Delta t} + \frac{V_{dc}}{R}$$

$$i_{Davg} = \frac{100\mu(1.13)}{23.5\mu} + \frac{10.735}{100}$$

$$i_{Davg} = 4.92 \text{ [A]}$$

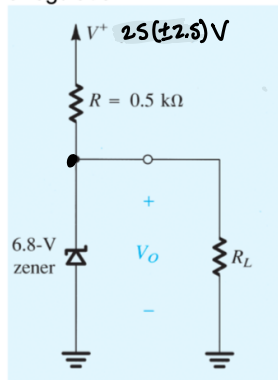
average diode current in FB

$$i_{Dmax} = 4.92 \text{ [A]}$$

since $i_{Lmax} \approx i_{Lavg}$ in Δt

2. Zener Diode Regulator: For this problem you will need the data sheet for a Zener Diode, which you can find your own or use the one Fairchild Semiconductor that is provided on Canvas. Note that the data sheet provides values for V_Z , which is measured at the current I_Z , as well as for the series resistance. We want to design a 12 V regulator using a reverse-bias Zener diode similar to the circuit below the load of 2.2 k Ω . For this design problem, the supply voltage V^+ is 25 ± 2.5 V (not 10 ± 1 V as shown in the figure below).

- Choose one Zener diode to regulate the load voltage to approximately 12 V. Based on the information in the data sheet, find V_{Z0} for this diode.
- If the Zener current is 25 mA, find V_Z . Then find the current through R , and from that determine a value for R . Use the nominal voltage for the power supply (i.e., 25 V).
- Find the line regulation.



$$V_{Z0} = V_Z - I_Z \cdot Z_Z$$

$$V_{Z0} = 12 - (.005)(20)$$

$$V_{Z0} = 11.9 \text{ [V]}$$

$$i_L = \frac{V_Z}{R_L} = \frac{12.4}{2.2k}$$

$$i_L = 5.64 \text{ [mA]}$$

$$i_R = i_Z + i_L$$

$$i_R = 25 + 5.64$$

$$i_R = 30.64 \text{ [mA]}$$

$$V_Z(25\text{mA}) = V_{Z0} + Z_Z I_Z$$

$$= 11.9 + (20)(.025)$$

$$V_Z(25\text{mA}) = 12.4 \text{ [V]}$$

Fairchild BZX55C2V4

$$V_Z(\text{nominal}) = 12 \text{ [V]}$$

$$I_Z = 5 \text{ [mA]}$$

$$Z_Z = 20 \text{ [}\Omega\text{]} = r_z$$

line regulation:

$$\Delta V_O = \Delta V^+ \left(\frac{r_z || R_L}{R + (r_z || R_L)} \right)$$

$$\frac{\Delta V_O}{\Delta V^+} = \frac{20 || 2.2k}{411 + (20 || 2.2k)}$$

$$\frac{\Delta V_O}{\Delta V^+} = 0.046$$

$$\Delta V_O = \text{line reg.} (\Delta V^+)$$

$$V^+ - V_Z = i_R \cdot R \quad \Delta V_O = (.046)(2.5)$$

$$R = \frac{V^+ - V_Z}{i_R}$$

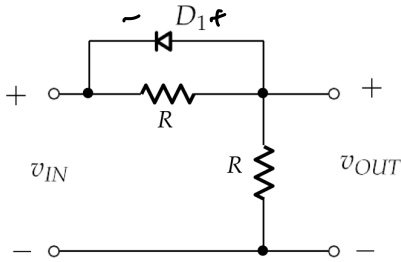
$$R = \frac{25 - 12.4}{30.64}$$

$$R = 411.3 \approx 411 \text{ [}\Omega\text{]}$$

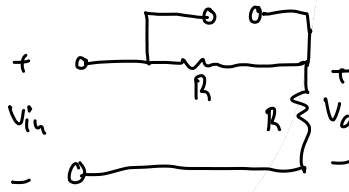
$$\Delta V_O = 0.115 \text{ [V]}$$

3. For the circuit below using the constant voltage drop model with $V_{D0} = 0.7\text{V}$ and $r_D = 0$:

- Write an expression for V_{OUT} in terms of V_{IN} .
- Sketch a graph of V_{OUT} vs. V_{IN} .



$$V_{in} = 5\text{mV}$$



$$V_O = V_{in} \left(\frac{R}{R+R} \right)$$

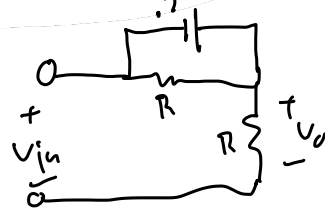
$$V_O = \frac{V_{in}}{2}$$

D_1 conducts only for $V_{out} \geq V_{in} + 0.7\text{ [V]}$

$$-0.7 \geq V_{in} - \frac{V_{in}}{2}$$

$$-0.7 \geq \frac{1}{2} V_{in}$$

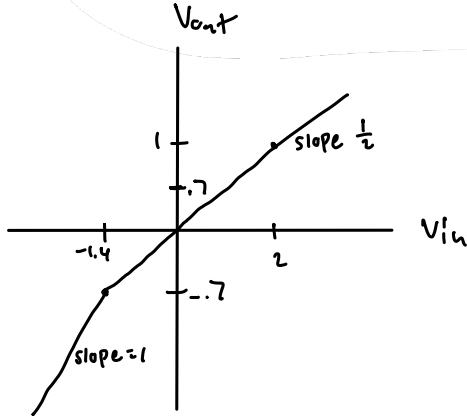
$$-1.4 \geq V_{in}$$



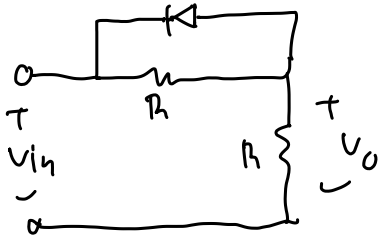
$$V_O - V_{in} = 0.7$$

$$V_O = 0.7 + V_{in}$$

$$V_O = \begin{cases} \frac{1}{2} V_{in}, & V_{in} \geq -1.4 \\ 0.7 + V_{in}, & V_{in} \leq -1.4 \end{cases}$$



4. Repeat problem 3 with $r_D = R$.



$$V_{in} \text{ small} \Rightarrow V_O = \frac{1}{2} V_{in}$$

$$V_O = \begin{cases} \frac{1}{2} V_{in}, & V_{in} \geq -1.4 \\ \frac{7}{30} + \frac{2}{3} V_{in}, & V_{in} \leq -1.4 \end{cases}$$

conducts for $V_{out} \geq V_{in} + 0.7$

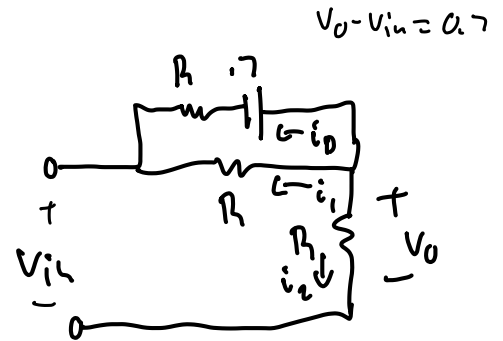
$$i_D + i_1 + i_2 = 0$$

$$\frac{V_O + 0.7 - V_{in}}{R} + \frac{V_O - V_{in}}{R} + \frac{V_O}{R} = 0$$

$$3V_O - 2V_{in} - 0.7 = 0$$

$$3V_O = 0.7 + 2V_{in}$$

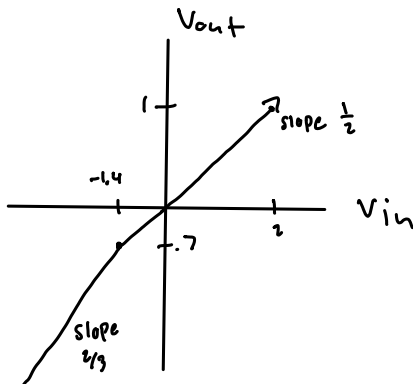
$$V_O = \frac{7}{30} + \frac{2}{3} V_{in}$$



$$\frac{7}{30} + \frac{2}{3} V_{in} \geq V_{in} + 0.7$$

$$-\frac{7}{15} \geq \frac{1}{3} V_{in}$$

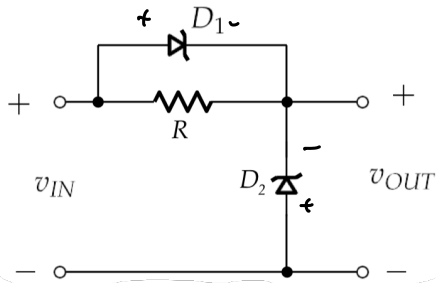
$$-1.4 \geq V_{in}$$



5. For the circuit below using the constant voltage drop model with $V_{D0} = 0.7\text{ V}$, $V_{Z0} = 1.4\text{ V}$,

$$r_D = r_Z = 0:$$

- Write an expression for V_{OUT} in terms of V_{IN} .
- Sketch a graph of V_{OUT} vs. V_{IN} .



$$V_{out} \leq -0.7, D_2 \rightarrow FB \therefore V_{out} = -0.7\text{ (V)}$$

$$V_{out} \geq 1.4, D_2 \rightarrow \text{rev. break down} \therefore V_{out} = 1.4\text{ (V)}$$

$$\text{need } V_{out} \text{ for } -0.7 \leq V_{in} \leq 1.4$$

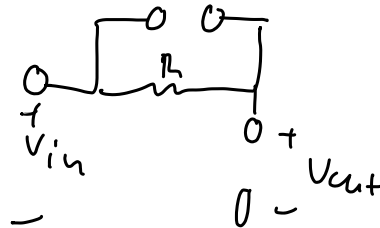
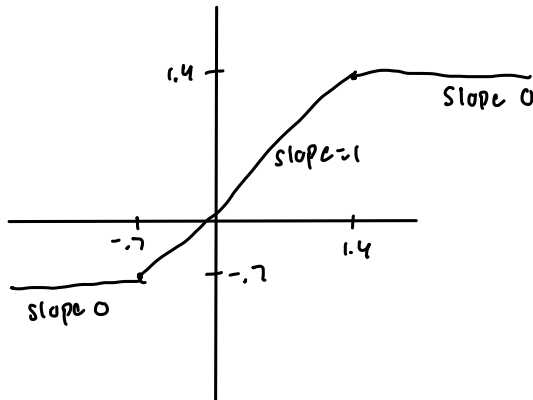
$$V_{in} - \text{small} : \text{cant overcome } V_{D0} = 0.7 \therefore D_1 \rightarrow \text{open}$$

$$V_{in} = 0.7, \text{ cant overcome}$$

$$V_{Z0} = 1.4 \text{ for } D_2 \rightarrow \text{open}$$

$$\therefore V_{out} = V_{in}$$

$$V_{out} = \begin{cases} 1.4\text{ (V)} & V_{in} \geq 1.4 \\ V_{in} & -0.7 \leq V_{in} \leq 1.4 \\ -0.7 & V_{in} \leq -0.7 \end{cases}$$



6. Measurements of V_{BE} and two terminal currents taken on a number of npn transistors operating in the active mode are tabulated below. For each, calculate the missing current value as well as α , β , and I_S as indicated by the table.

Transistor	a	b	c	d	e
V_{BE} (mV)	700	690	580	780	820
I_C (mA)	1.000	1.000	0.23	10.10	73.45
I_B (μA)	10	20	5	120	1050
I_E (mA)	1.01	1.020	0.235	10.22	75.00
α	0.99	0.98	0.979	0.988	0.986
β	100	50	46	84	70.43
I_S	$6.91 \cdot 10^{-16}$	$1.03 \cdot 10^{-15}$	$1.93 \cdot 10^{-14}$	$2.85 \cdot 10^{-16}$	$4.21 \cdot 10^{-16}$

$$I_E = I_B + I_C \quad \alpha = \frac{I_C}{I_E}$$

$$\beta = \frac{\alpha}{1-\alpha}$$

$$I_C = I_S e^{\frac{V_{BE}}{V_T}}$$

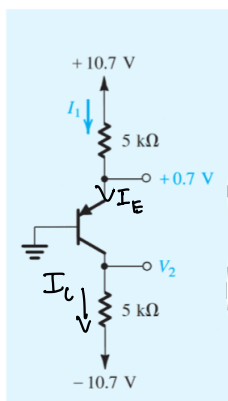
$$V_T \approx 25\text{ mV}$$

$$I_m = I_S e^{\frac{700\text{ mV}}{25\text{ mV}}}$$

$$I_m e^{-28} = I_S$$

$$I_S = 6.91 \cdot 10^{-16}$$

7. For the circuits below, assume that the transistors have very large β . Some measurements have been made on these circuits, with the results indicated in the figure. Find the values of the other labeled voltages and currents.



$$I_1 = \frac{10.7 - 0.7}{5 \text{ k}} = 2 \text{ mA}$$

$$I_1 = 2 \text{ mA} = I_E$$

$$I_C = \beta I_B \Rightarrow I_B = \frac{I_C}{\beta}$$

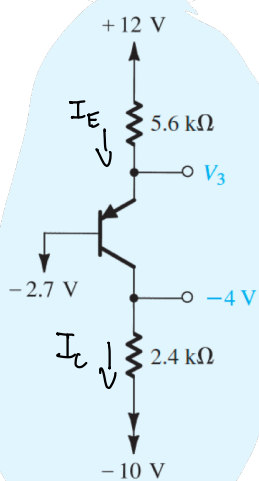
$$I_E = I_C + I_B \Rightarrow I_C + \frac{I_C}{\beta}$$

$$I_E = I_C \left(1 + \frac{1}{\beta}\right) \quad \beta \rightarrow \infty \Rightarrow I_E \approx I_C \therefore I_C \approx 2 \text{ mA}$$

$$V_2 - (-10.7) = I_C \cdot 5 \text{ k}$$

$$V_2 = I_C(5 \text{ k}) - 10.7$$

$$V_2 = -0.7 \text{ V}$$



$$I_C \approx I_E \text{ since } \beta \rightarrow \infty$$

$$I_E \approx I_C = \frac{-4 - (-10)}{2.4 \text{ k}} = 2.5 \text{ mA}$$

$$\frac{12 - V_3}{5.6 \text{ k}} = I_E$$

$$12 - I_E(5.6 \text{ k}) = V_3$$

$$V_3 = -2 \text{ V}$$

$$I_C \approx I_E, I_B \approx 0$$

$$I_C + I_B = I_E$$

$$I_C = I_E = I_Q$$

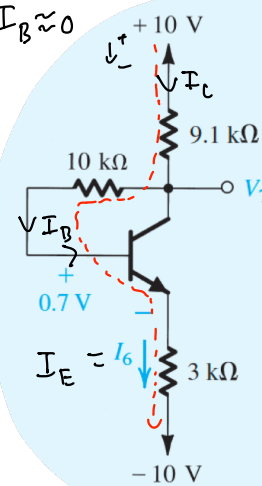
KVL:

$$10 + I_C(9.1 \text{ k}) + I_B(10 \text{ k}) + 7 + I_C(3 \text{ k}) - (-10) = 0$$

$$I_C(9.1 \text{ k} + 3 \text{ k}) = 20 - 7$$

$$I_C = \frac{13}{12.1 \text{ k}}$$

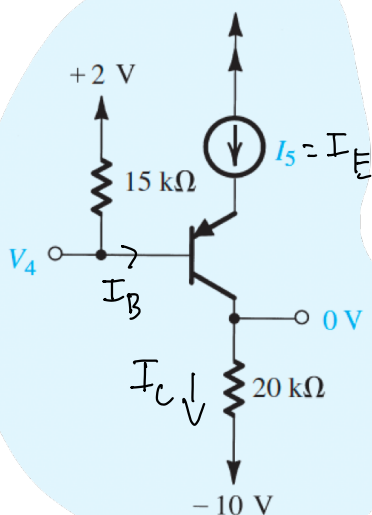
$$I_C = 1.545 \text{ mA}$$



$$\frac{10 - V_7}{9.1 \text{ k}} = I_C$$

$$10 - I_C(9.1 \text{ k}) = V_7$$

$$V_7 = -4.515$$



$$I_E \approx I_C = \frac{0 - (-10)}{20 \text{ k}} = 0.5 \text{ mA} = I_S$$

$$I_B = \frac{I_C}{\beta}, \quad \beta \rightarrow \infty \Rightarrow I_B = 0$$

$$\frac{2 - V_4}{15 \text{ k}} = 0$$

$$V_4 = 2 \text{ V}$$